

ECO362 Final Solutions

Narek Alexanian, Stephanie Hu, Jiwon Lee, Jialiang Lin

December 31, 2023

Q1

- (a) False. If long-term bonds have a diversification effect, i.e., decreasing the portfolio risk, then the investor might want to invest in them despite lower expected returns.
- (b) True. See p.6 and p.11 of the Alpha lecture slides.
- (c) False. After captured by SMB and HML factors, that strategy might be still able to earn positive alpha, for example, by picking the right small-cap growth stocks.
- (d) False. The information ratio is a better measure since it uses relevant benchmark returns.
- (e) False. If you are an expected mean-variance utility maximizer, then the optimal amount of insurance is equal to your human capital regardless of your risk-aversion.
- (f) True. The pension fund has long-term, fixed liabilities and relatively short-term, floating assets. Thus, in order to hedge this risk, it should enter the interest rate swap that receives fixed and pays floating.

Grading:

2 pt: Correct answer. No partial credit.

Q2

- (a) Verizon is a value stock. (Value stocks are those with relatively low price-dividend ratios.)

Grading:

4 pt: Correct answer.

- (b) Microsoft is a growth stock. (Growth stocks are those with relatively high price-dividend ratios.)

Grading:

4 pt: Correct answer.

- (c) Apply CAPM, $\mathbb{E}[r_i - r_f] = \beta \mathbb{E}[r_m - r_f]$, to Bank of America and Coca-Cola:

$$\begin{aligned} 21.46\% - r_f &= 1.68 \mathbb{E}[r_m - r_f], \\ 9.40\% - r_f &= 0.41 \mathbb{E}[r_m - r_f] \end{aligned} \tag{1}$$

and one can solve for $\mathbb{E}[r_m - r_f] = 9.50\%$ and $r_f = 5.51\%$. So the annual riskless interest rate is 5.51%.

Grading:

4 pt: Correct equations for which the riskless interest rate must satisfy and correct numerical results.

2 pt: Correct equations for which the riskless interest rate must satisfy.

- (d) Apply CAPM to Microsoft:

$$\mathbb{E}[r - 5.51\%] = 0.88 \times 9.50\%, \tag{2}$$

which yields $\mathbb{E}[r] = 13.86\%$. So the expected annual return for Microsoft is 13.86%.

Grading:

4 pt: Correct equation for which the expected return must satisfy and correct numerical results.

2 pt: Correct equation for which the riskless interest rate must satisfy.

- (e) Apply CAPM to Verizon:

$$\mathbb{E}[r - 5.51\%] = 0.24 \times 9.50\%, \tag{3}$$

which yields $\mathbb{E}[r] = 7.79\%$.

Apply the Gordon growth model:

$$14 = \frac{1 + G}{7.79\% - G}, \tag{4}$$

which yields $G = 0.60\%$. So the expected annual dividend growth for Verizon is 0.60%.

Grading:

4 pt: Correct equations for which the expected return and dividend growth rate must satisfy and correct numerical results.

3 pt: Correct equations for which the expected return and dividend growth rate must satisfy.

2 pt: At least one correct equation for which either the riskless interest rate or dividend growth rate must satisfy.

Q3

- (a) Stocks that are *cheap* are better approximated by stocks with high market-to-book equity ratios (you pay the least in terms of market price per unit of book-value of the firm). Thus, the investor's strategy is best approximated by the value strategy which results in a wealth at 1945 equal to:

516.31\$

Grading:

6 pt: Correct answer

5 pt: Numerical error, but correct formula

2pt: Answer value

- (b) By CAPM we know that for each year and portfolio i,

$$R_{i,t} - R_{b,t} = \alpha_i + \beta_i \cdot (R_{m,t} - R_{b,t}) + \epsilon_{i,t}$$

Then we can calculate, $\alpha_i + \epsilon_{i,t}$ for each year from 1940 to 1945. Since $R_{b,t} = 0$ for each year:

- $\alpha_i + \epsilon_{i,1940} = -15.5\%$
- $\alpha_i + \epsilon_{i,1941} = 16\%$
- $\alpha_i + \epsilon_{i,1942} = 28\%$
- $\alpha_i + \epsilon_{i,1943} = 20\%$
- $\alpha_i + \epsilon_{i,1944} = 37.5\%$
- $\alpha_i + \epsilon_{i,1945} = 7.5\%$

Then, since the expected error is zero,

$$\alpha_i = \frac{-15.5 + 16 + 28 + 20 + 37.5 + 7.5}{6}\% = 15.58\%$$

equals the average of the values calculated above.

Grading:

6 pt: Correct answer

5 pt: Numerical error(ex. missing percentage sign)

4 pt: Find single alpha through geometric averaging

3 pt: Simple average of cumulative return; mixing arithmetic and geometric averaging / annualized return over 5 years

2 pt: Calculate alpha over cumulative returns without any averaging; calculate $\alpha + \epsilon$ for an arbitrary chosen year

0pt: Simply stating CAPM equation

- (c) The average annual return of the market over the sub-period 1940-1945 is 14.5% which is 2.5% higher compared to long-run average of 12%. Hence, the component of return that comes from the over-performance of market during the sub-period amounts to $1.5 * 2.5 \%$, where $\beta = 1.5$. This is a small part of the total average return of 37.33% of the value strategy which is mostly attributed to alpha or *stock-picking*.

Grading:

4 pt: Correct conclusion by calculating and comparing the average returns of the market in the sub-period and mentioning positive alpha

2 pt: Correct answer but missing comparison between sub-period market return and long-run return and through alpha due to CAPM

0.5 pt: Correct answer(stock picking); incorrect explanation

Q4

- (a) $H = 2 \sum_{t=1}^{10} P_b(t) + 68 \sum_{t=11}^{20} P_b(t) = 2 \times 7.98 + 68 \times (13.02 - 7.98) = 15.96 + 342.72 = 358.68$ (in million dollars).

Grading:

4 pt: Correct equation and answer

2 pt: Correct equation but errors in calculation and answer

1 pt: Fair approach but chose wrong annuity prices or annual income

0 pt: Didn't understand the concept of annuity

- (b) $\frac{S}{A} = \left(1 + \frac{H}{A}\right) \frac{\mu - r_b}{\gamma \sigma^2}$. The one-year riskless return is $r_b = 1/0.95 - 1 = 5.26\%$, assuming it is annually compounded. Then, $\frac{S}{A} = \left(1 + \frac{358.68}{100}\right) \frac{12\% - 5.26\%}{10 \times 0.2^2} = 77.25\%$.

Grading:

4 pt: Correct equation and answer

2 pt: Correct equation but errors in calculation and answer

2 pt: Correct equation for the stock weight but wrong riskless return or human capital (e.g., use a different value from 4(a))

1 pt: Correct equation for the stock weight but wrong riskless return and human capital

0 pt: Wrong equation (e.g., didn't include human capital)

- (c) In the first scenario, after-tax human capital $H' = H \times (1 - 0.51) = 358.68 \times (1 - 0.51) = 175.75$. In the second scenario, $H' = 15.96 \times (1 - 0.51) + 342.72 \times (1 - 0.37) = 223.73$ (in million dollars).

Grading:

4 pt: Correct equations and answers for both scenarios

2 pt: Correct equation and answer for one scenario

(-1 pt per calculation errors in one scenario)

0 pt: Incorrect

- (d) No. This contract does not protect Ohtani from inflation or interest rate risk. There is also a chance that LA Dodgers will go bankrupt, which means the contract is exposed to default risk. Moreover, Human capital is exposed to the mortality risk (i.e, the risk of death). In that case, it is optimal to hedge it with life insurance.

Grading:

4 pt: Correct answer and explanation

2 pt: Correct answer but wrong explanation

0 pt: Correct answer (Y/N) without explanation

Q5

- (a) $\bar{y}$ is the long-run average of short-run yields

μ is the term-premium which equals the expected excess return of any long-term bond compared to one-period short bond, $\mu = \mathbb{E} [r_1(t) - y_0(1)]$

To calculate the value we solve the system of equations(in percentages):

$$y(2) = \bar{y} + \frac{1}{2} \cdot \frac{1 - 0.8^2}{1 - 0.8} \cdot (y(1) - \bar{y}) + \frac{\mu}{2}$$

$$y(3) = \bar{y} + \frac{1}{3} \cdot \frac{1 - 0.8^3}{1 - 0.8} \cdot (y(1) - \bar{y}) + \frac{2 \cdot \mu}{3}$$

The system yields $\bar{y} = 2.88\%$ and $\mu = 0.33\%$

Grading:

+4 pt: Correct calculation/set-up and definition

+3 pt: Numerical error, but correct set-up of system of equations / incorrect definition, ex. write expected short-rate instead of long-run average(note difference between unconditional and conditional mean)

+1 pt: Correct definition

- (b) See above

Grading:

+4 pt: Correct calculation/set-up and definition

+3 pt: Numerical error, but correct set-up of system of equations / incorrect definition

+1 pt: Correct definition

- (c) Since $y_0(1) < \bar{y}$, hence we expect the short-rate in future to increase by mean-reversion.

Grading:

+4 pt: Correct explanation, mentioning mean-reversion

+3 pt: If justify answer with upward slope;(upward slope maybe due to positive term-premium and is about long-term yields not future yields)

+2 pt: Correct conclusion by comparing the long-run average to the current short-rate but not mention mean-reversion assumption.

- (d) As t get bigger and bigger the second term converges to 0, while the third term converges to μ ; hence,

$$y(\infty) = \bar{y} + \mu = 3.2\%$$

Grading:

+4 pt: Correct equation; carry-over numerical errors not penalized

+0 pt: Incorrect answer or calculation for arbitrarily chosen large t

(e) By the expectation hypothesis, the return of any long-term bond is:

$$\mathbb{E}[r_1(3)] = \mu + y_0(1) = 2.33\%$$

Grading:

+4 pt: Correct equation; either by directly using the expectation hypothesis or calculate the expected value of $y_1(2)$ by first calculating expected $y_1(1)$ through mean reversion (carry-over numerical errors not penalized)

+3 pt: Correct set-up but numerical error

+1 pt: Proxy $y_1(2)$ through $y_0(2)$ or the forward rate $y_f(1,3)$ and then calculate return under continuous compounding

Note: Wrong methods can yield answers that are numerically close to the correct one, but still be penalized according to the rubric above.

Q6

- (a) Under continuous compounding, the futures price consistent with no arbitrage is $F = S_0 e^{r_b T}$. The spot price (S_0) is 466, the annual riskless rate (r_b) is 5%, and the maturity of the futures is 4 months, so $T = \frac{1}{3}$. Solving for the futures price:

$$F = S_0 e^{r_b T} = 466 \cdot e^{\frac{0.05}{3}} \approx 473.83$$

Grading:

- 4 pt: Correct setup and answer
 - 3 pt: Correct setup (numbers are correct in the equation) but incorrect final answer
 - 2 pt: Up to one error in setup
 - 1 pt: Attempt with annual compounding
 - 1 pt: Correct formula
- (b) To compute the put option price under no arbitrage, we use put-call parity. Given the call option price C , the put price is $P = C - S_0 + X e^{-r_b T}$. C is \$34, S_0 is 466, r_b is 5%, T is $\frac{1}{6}$ (since the put has a 2 month maturity and the riskless rate is annual), and the exercise price is \$460. Solving for the put price:

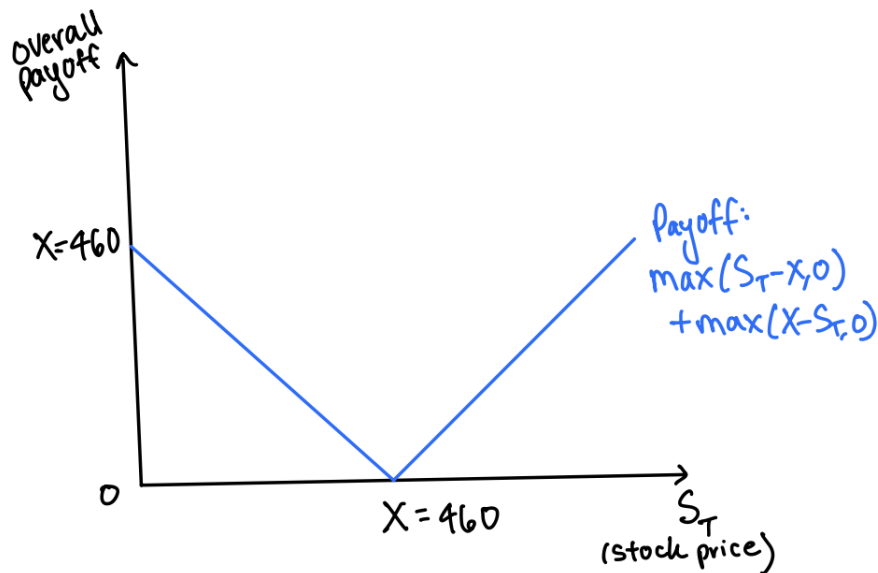
$$P = C - S_0 + X e^{-r_b T} = 34 - 466 + 460 \cdot e^{-\frac{0.05}{6}} \approx 24.18$$

Grading:

- 4 pt: Correct setup and answer
 - 3 pt: Correct set up but incorrect final answer
 - 2 pt: Up to one error in setup
 - 1 pt: Correct formula
- (c) The long straddle strategy is as follows:
1. Long/buy a call at price C with exercise price X and time to expiration T
 2. Long/buy a put at price P with exercise price X and time to expiration T

The overall payoff at expiration is the sum of the call payoff and put payoff, so it is $\max(S_T - X, 0) + \max(X - S_T, 0)$.

Below is the payoff diagram for the long strategy:



Grading:

- +0.5 pt: Correctly label y-axis intercept
- +0.5 pt: Correctly label x-axis intercept
- +0.5 pt: Correctly provide payoff expression
- +1.5 pt: Correctly draw graph of payoff diagram
+1 pt: Correctly draw graph of profit diagram, or draw any graph with correct shape but minimum point is not on the x-axis
- +1 pt: Correctly describe straddle strategy
- Note that answers based on the short straddle strategy are accepted and graded in the same way.

- (d) The profit expression for the straddle strategy is $\max(S_T - X, 0) + \max(X - S_T, 0) - (P + C)$. In order to break even, we set the profit to 0, so $\max(S_T - X, 0) + \max(X - S_T, 0) = P + C$, where $P + C \approx 24.18 + 34 \approx 58.18$. Since the stock price today is \$466, the stock price will need to move by at least $58.18 - 6 = 52.18$ to the right or at least $58.18 + 6 = 64.18$ to the left to break even.

The straddle strategy is betting on stock price volatility. The long position bets on high volatility (and the short position bets on low volatility).

Grading:

- +3 pt: Correct answer for stock price movement to break even
- +2 pt: States stock price needs to move in either direction by $P+C$ without taking into account original stock price
- +1 pt: Correct answer for what straddle strategy bets on
- Carry-over errors for put price not penalized